

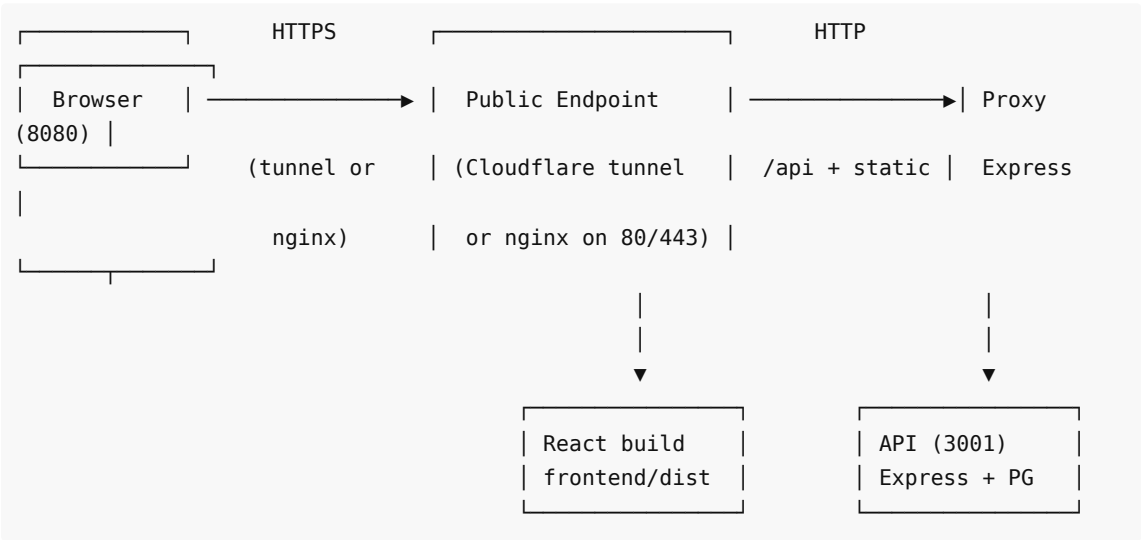
VMA Calculator Service Overview

1. What the service does

The VMA Calculator helps coaches and athletes capture endurance test results (Cooper and Demi-Cooper), compute VMA metrics, and store authenticated users' histories. It ships with:

- **Public SPA** (React + Vite + Tailwind) for running guest tests or logged-in workflows.
- **Secure API** (Node.js + Express + PostgreSQL) with JWT authentication, password reset, and per-user data isolation.
- **Email hooks** (Mailjet) for welcome and password reset notifications (currently pending Mailjet account unblock).

2. High-level architecture



Key pieces:

Component	Role	Location
Frontend SPA	Built assets served by <code>scripts/proxy.js</code> (port 8080)	<code>frontend/</code> + <code>frontend/dist</code>
API server	Auth, tests CRUD, per-user stats	<code>backend/src</code> (port 3001)
PostgreSQL	Persistent storage (users, tests, reset tokens)	Local DB <code>vma_calculator</code>
Proxy	Serves SPA and rewrites <code>/api</code> → backend	<code>scripts/proxy.js</code>
Public access	Either trycloudflare tunnel (<code>scripts/start_tunnel.sh</code>) or future nginx/Caddy on 80/443	<code>scripts/</code> , system services

3. Deployment modes

3.1 Local-only

1. `cd backend && npm install` (once).
2. `npm run dev` or `node src/index.js` (port 3001).
3. `cd frontend && npm install && npm run dev` (Vite dev server) or run `npm run build` and serve through proxy.

3.2 One-command public demo (Cloudflare quick tunnel)

Run the orchestrator from the repo root:

```
./scripts/start_tunnel.sh
```

What it does:

1. Builds the frontend (Vite → `frontend/dist`).
2. Starts the backend if it is not already running (background, logs → `backend.log`).
3. Installs proxy dependencies in `scripts/` and launches `node scripts/proxy.js` (logs → `proxy.log`).
4. Installs `cloudflared` if missing, then starts a quick tunnel that exposes `http://localhost:8080` and prints a URL like `https://your-tunnel.trycloudflare.com` .

Share that URL with external testers; the SPA will call `/api/*` relative to the same origin, so both UI and API work through a single hostname. Keep the script session running—closing it stops the tunnel.

3.3 Direct hosting from your machine (recommended for stability)

1. Keep backend + proxy running permanently (PM2/systemd).
2. Install nginx or Caddy, pointing upstream to `http://127.0.0.1:8080` .
3. Forward router ports 80/443 to this machine, and point DNS (A record) to your public IP.
4. Use Certbot or Caddy automatic TLS so you serve HTTPS directly.

4. Operations & monitoring

- **Logs** (all relative to repo root):
 - Backend API: `tail -f backend.log`
 - Proxy server: `tail -f proxy.log`
 - Cloudflare tunnel: `tail -f cloudflared.log`
 - Nginx (when enabled): `sudo journalctl -u nginx -f`
- **Health checks:**
 - Direct API: `curl http://localhost:3001/api/health`
 - Through proxy: `curl http://localhost:8080/api/health`
 - Public URL: `curl https://<tunnel-or-domain>/api/health`
- **Database:** `psql -h localhost -U postgres -d vma_calculator -c "SELECT ..."`
- **Logs inside PostgreSQL:** use `\dt` , `\d tests` , etc., from `psql` to inspect schema.

5. Accounts & secrets

- `backend/.env` controls DB credentials, JWT secret, and Mailjet keys. Keep this file out of version control.
- Production recommendations:
 - Rotate `JWT_SECRET` and DB credentials before exposing publicly.

- Enable SSL on PostgreSQL if remote clients will access it.
- Store secrets in a password manager or `.env` managed by your process manager.

6. Known limitations / next steps

1. **Mailjet blocked:** Welcome/reset emails currently fail with HTTP 401 from Mailjet. Resolve with Mailjet support or switch to another provider (SendGrid, SES, Postmark) by updating `emailService.js` and `.env`.
2. **Availability:** Quick tunnels are temporary. For demos, consider a named Cloudflare tunnel with DNS mapping or direct nginx hosting.
3. **Process management:** For continuous uptime, add PM2/systemd units for both backend and proxy, plus a named tunnel or nginx service.
4. **Monitoring:** Add uptime checks (e.g., health ping monitors) and log rotation.

7. Quick reference commands

```
# Start everything via tunnel script
./scripts/start_tunnel.sh

# Stop proxy/backend/tunnel manually
pkill -f "scripts/proxy.js"
pkill -f "node src/index.js"
pkill -f cloudflared

# View logs
tail -f backend.log proxy.log cloudflared.log

# Database peek
PGPASSWORD='VmaCalc2026!' psql -h localhost -U postgres -d vma_calculator -c 'SELECT
COUNT(*) FROM tests;'
```

8. Contact / ownership

- **Primary maintainer:** gladius (this machine/user).
- **Tech stack:** React 18, Vite, Tailwind, Node 18, Express, PostgreSQL, Mailjet, Cloudflare Tunnel.
- **Source repository:** local workspace `/home/gladius/Codes/project`.

This document should accompany the generated PDF for stakeholders who need an offline summary of the current deployment setup.